01 | What our dataset contains

Automated successful expert demonstrations, not learned-policy successes

Episode	Use	Commands	Simulated time	Expert outcome
train_00000	train	1,792	35.83 s	success
train_00001	train	1,807	36.13 s	success
train_00002	train	1,902	38.03 s	success
train_00003	validation	1,750	34.99 s	success

Observation at time t

Fixed RGB + wrist RGB: 320 x 240
16 robot joint/gripper values
Instruction: insert the plug

Expert command at time t

7 right-arm joint targets [rad]
1 per-finger travel target [m]
Requested + applied values saved

Timing and provenance

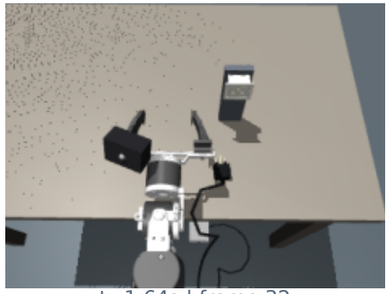
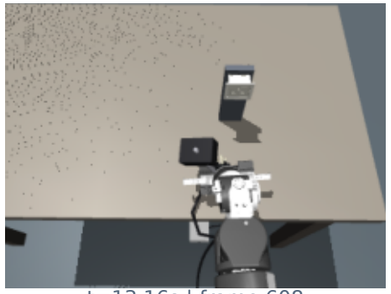
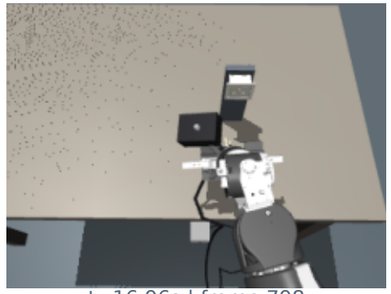
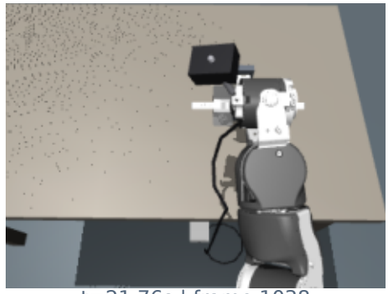
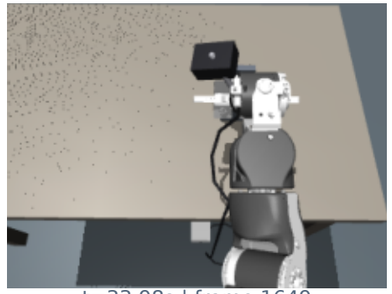
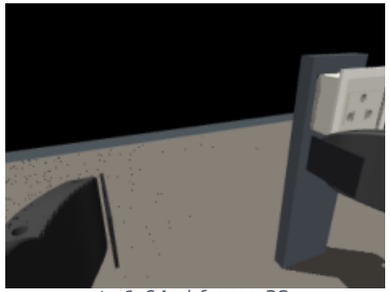
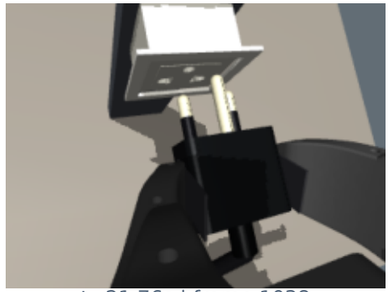
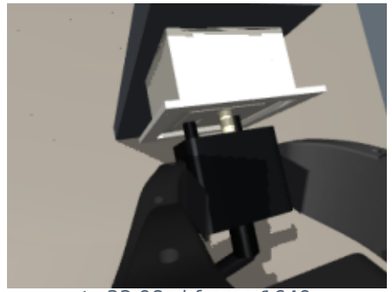
50 Hz command boundaries
T commands, T+1 observations
Seeds, phases, hashes, outcomes

7,251 total transitions | 5,501 train / 1,750 validation

Scope: four locally available episodes; explicit split override. Missing train_00005 is excluded, not fabricated. Source: raw NPZ + metadata.

02 | What the model sees

Actual recorded RGB frames from train_00000 (training episode)

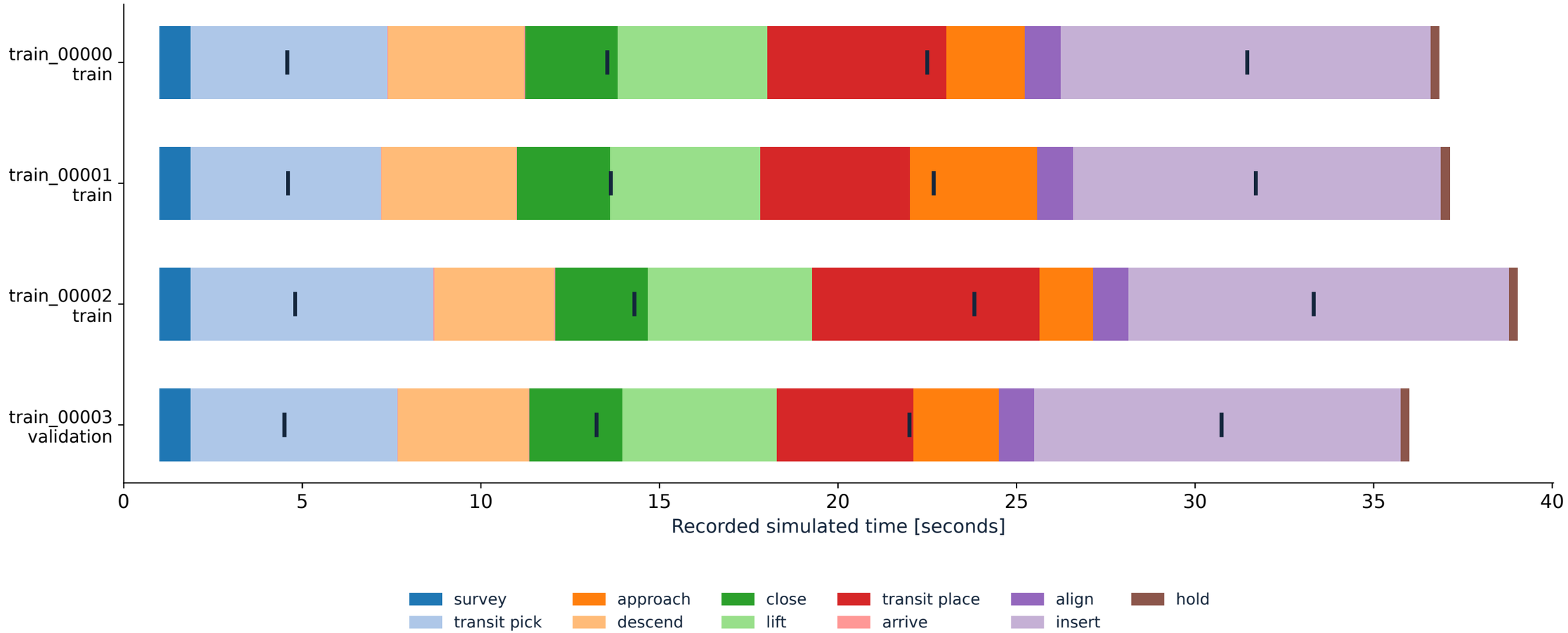
	Table start	Reach / descend	Grasp / close	Lift	Transport	Insert
Fixed camera	 t=1.64s | frame 32	 t=10.24s | frame 462	 t=13.16s | frame 608	 t=16.96s | frame 798	 t=21.76s | frame 1038	 t=33.98s | frame 1649
Wrist camera	 t=1.64s | frame 32	 t=10.24s | frame 462	 t=13.16s | frame 608	 t=16.96s | frame 798	 t=21.76s | frame 1038	 t=33.98s | frame 1649

Frames show expert behavior. Phase labels are presentation annotations, never actor inputs.

Lossless source pixels; resized for the slide only. The instruction and robot state accompany both images.

03 | Full episodes vs. the tiny training test

Colored segments: recorded expert phases. Black ticks: selected qualification window starts.



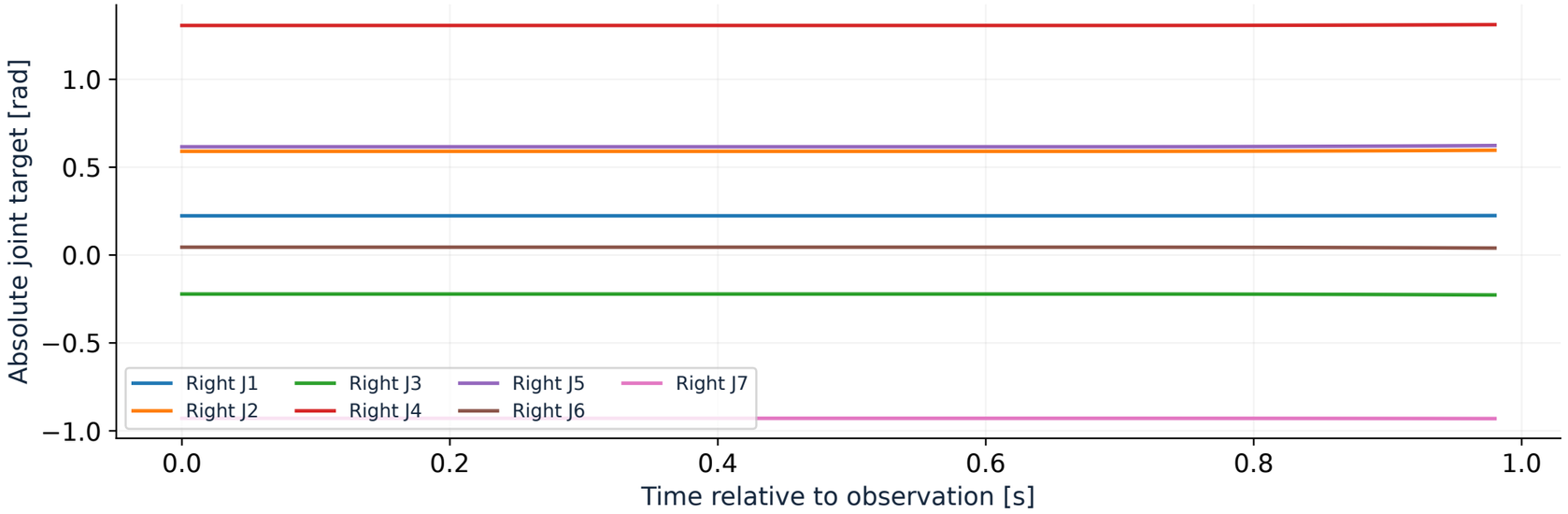
The export contains all frames; the 24-update qualification used only 12 train + 4 validation windows.

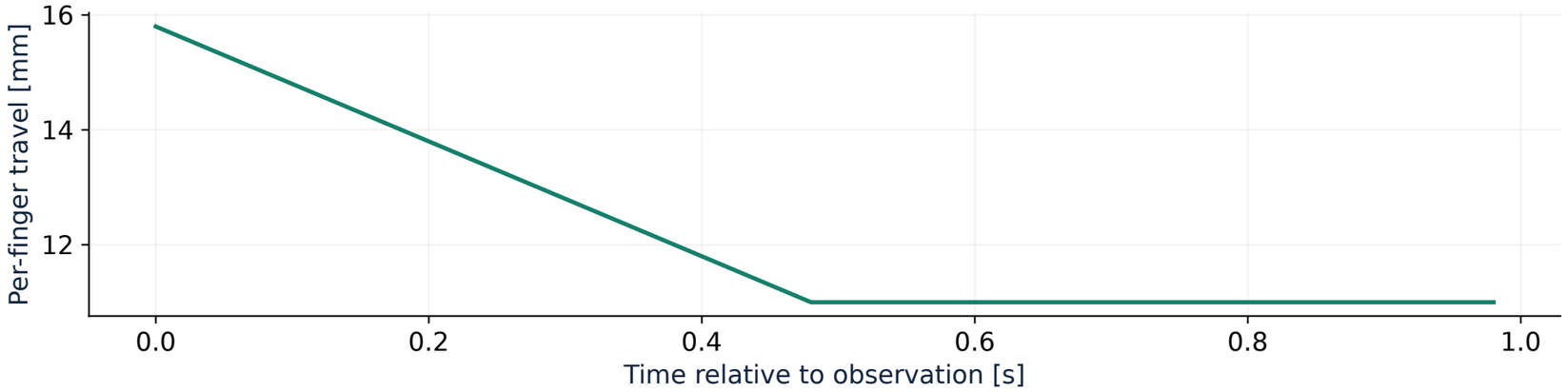
Next: train across the complete training episodes, including the initial survey/table-start behavior.

Source: phase and timestamp arrays + qualification_v2.json sample_plan. Validation ticks are evaluation only, not training updates.

04 | One training sample

Observation at t + instruction -> future applied expert targets (50 commands at nominal 50 Hz)





Actual robot state at t

Left: 7 joints + mean finger travel
Right: 7 joints + mean finger travel
16 values total; both arms observed

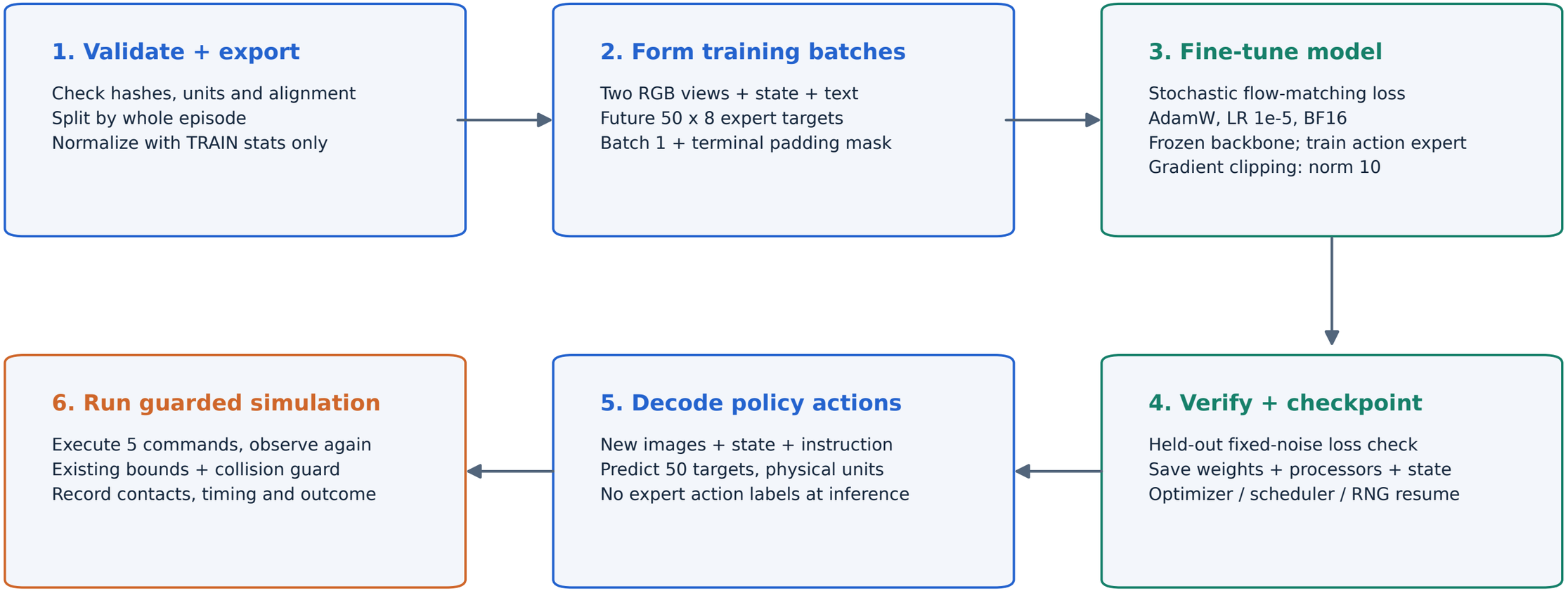
Right joints [rad]:
J1: 0.2235
J2: 0.5905
J3: -0.2217
J4: 1.3067
J5: 0.6170
J6: 0.0454
J7: -0.9289
Finger: 18.78 mm

Near episode end:
real actions contribute to loss;
padded future actions do not.
Last command: 1 valid + 49 padded.

Source: train_00000 step 608. Gripper values shown in mm for readability; stored/trained in metres. Labels are targets, not next-state positions.

05 | How we fine-tune and use SmolVLA

Implemented pipeline using pretrained SmolVLA and LeRobot 0.4.4 model/processors

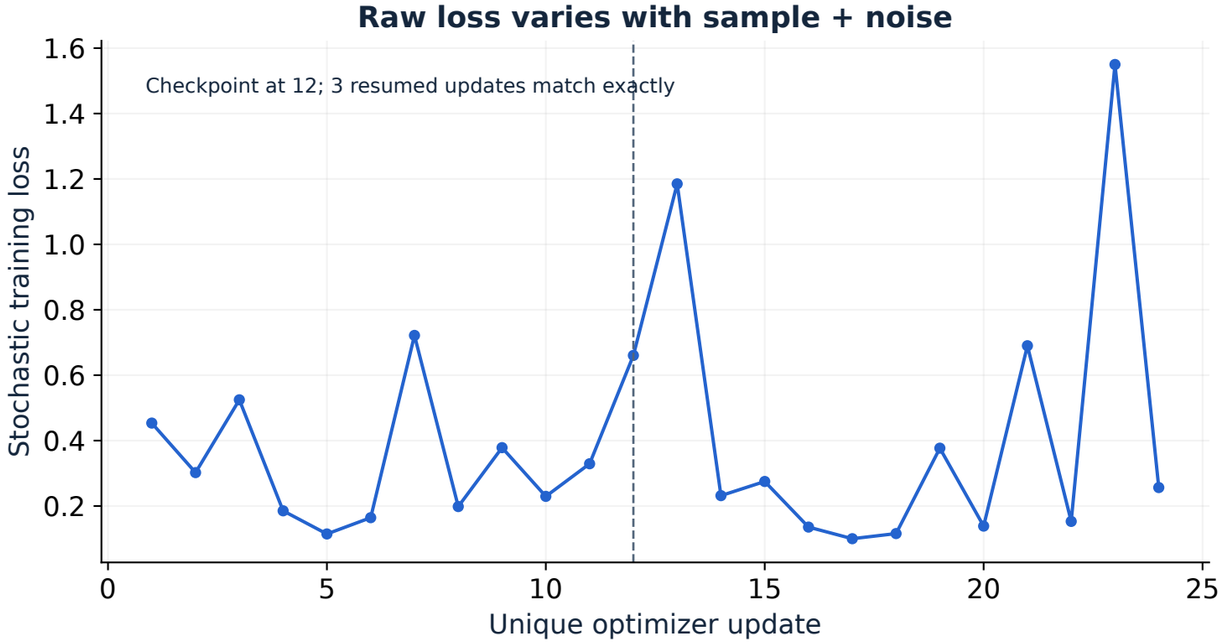


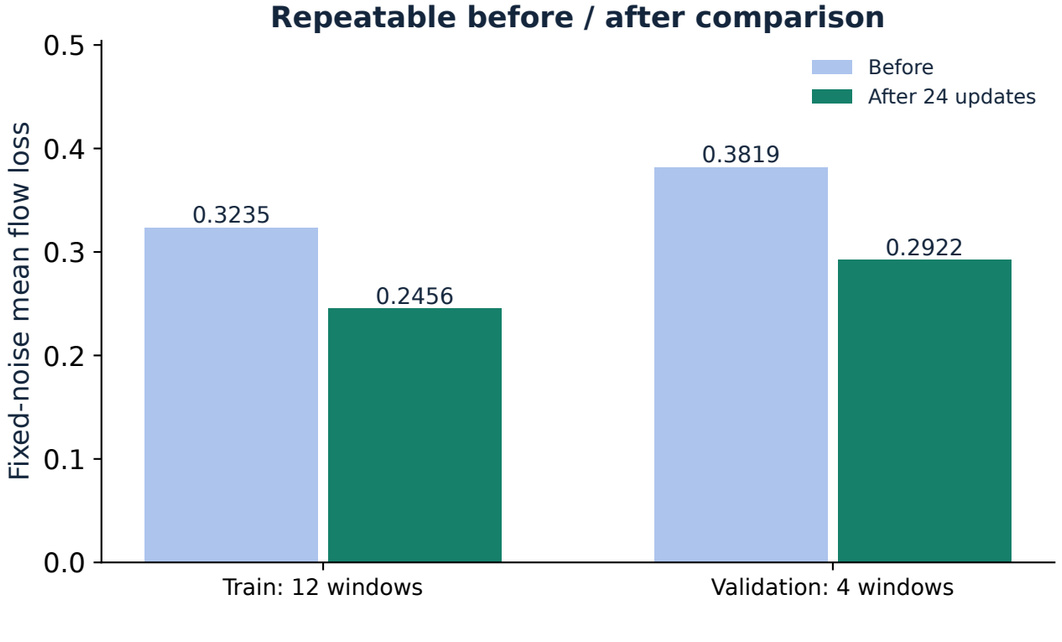
Privileged plug/socket poses and phase/contact diagnostics support the expert/evaluation only; never the VLA actor.

This diagram describes the implemented diagnostic scripts, not qualification of a long-run trainer, online RL, hardware deployment or successful learned insertion.

06 | Evidence that training executes

Actual saved test results: short-run numerical health and reproducibility, not autonomous task success





Measured: RTX 4060 Ti | batch 1 / BF16 | peak allocated 2.14 GB | 80.4s test interval

Finite losses/gradients; exact saved-state resume. Earlier smoke test verified exact model/processor reload.

Boundary: learned policy stopped after 12 commands, before grasp (robot_collision). No task-success claim.

Next: all-frame supervised training, longer stability tests, more demonstrations, then reserved-reset evaluation. RL comes later.

Sources: qualification_v2.json, smoke_v3.json, heldout_v2/report.json. Memory excludes other GPU usage; test timing includes checks, not imports or downloads.